LIBRARY BOOK TRACKER SYSTEM

A Project Report submitted

In partial fulfillment of the requirements for the award of the degree

of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE & ENGINEERING (DATA SCIENCE)

Submitted by

NAME – TARIKA SATYA

Reg. No. - 2025142284

Under the esteemed guidance of

Dr. Devarapalli Ravi Kishore, Associate Professor, CSE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

GITAM SCHOOL OF TECHNOLOGY

GITAM (Deemed to be University)

VISAKHAPATNAM

2025

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

GITAM SCHOOL OF TECHNOLOGY
GITAM (Deemed to be University)

DECLARATION

I hereby declare that the project report entitled **"LIBRARY BOOK TRACKER SYSTEM"** is an original work done in the Department of Computer Science and Engineering (Data Science), GITAM School of Technology, GITAM (Deemed to be University) submitted in partial fulfillment of the requirements for the award of the degree of B.Tech. in Computer Science and Engineering. The work has not been submitted to any other college or University for the award of any degree or diploma.

Date:

Registration No	Name	Student Signature
2025142284	TARIKA SATYA	

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(DATA SCIENCE)

GITAM SCHOOL OF TECHNOLOGY

GITAM (Deemed to be University)

CERTIFICATE

This is to certify that the project report entitled "LIBRARY BOOK TRACKER SYSTEM" is a bonafide record of work carried out by Tarika Satya Reg No. 2025142284, submitted in partial fulfillment of requirement for the award of degree of Bachelors of Technology in Computer Science and Engineering (Data Science).

Date :

Project Guide

Dr. Devarapalli Ravi Kishore

Associate Professor

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who have supported and guided me throughout the course of this project.

First and foremost, I extend my heartfelt thanks to **K. Suha Khubra, Shruti Saroj, Tangudu Akhilesh** for their invaluable assistance and unwavering support, which were instrumental in the successful completion of this project. Their extensive knowledge and profound insights greatly influenced the direction and outcome of my work.

I am deeply indebted to **Dr. Devarapalli Ravi Kishore** for his significant contributions and continuous support. His expertise and guidance were pivotal in overcoming the challenges encountered during various stages of this project.

Lastly, I would like to express my appreciation to my team members for their constant support and motivation. Their belief in my abilities gave me the strength to persevere and strive for excellence.

.

TABLE OF CONTENTS

S.No.	Description	Page No.
1.	Abstract	6
2.	Introduction	7
3.	Aim of the Project	8
4.	Problem Definition & Objectives	9
5.	Coding	10 – 13
6.	Output Screens	14
7.	Future Work	15
8.	Conclusion	16

ABSTRACT

In the modern era, efficient management of resources is essential in every field, including libraries. Traditional methods of maintaining library records manually are often time-consuming, error-prone, and difficult to manage. This project presents a Library Book Tracker System developed using the Python programming language to simplify and automate basic library operations.

The system allows users to perform essential tasks such as adding new books, viewing available books, issuing books to users, and returning books. It uses simple data structures like dictionaries to store and manage book information effectively.

This project is specifically designed for beginners to understand how programming concepts can be applied to solve real-world problems. It also helps in developing logical thinking, problem-solving skills, and a deeper understanding of Python fundamentals.

.

INTRODUCTION

A library is an important place where knowledge and information are stored in the form of books. Managing a library manually involves maintaining registers, tracking issued books, and ensuring proper record-keeping. This process can become difficult, especially when the number of books increases.

With the advancement of technology, computer-based systems have replaced manual systems to improve efficiency and accuracy. Python, being a simple and powerful programming language, is widely used for developing such systems.

The Library Book Tracker System is a basic implementation that demonstrates how programming can be used to manage data efficiently. The system is menu-driven and user-friendly, allowing even beginners to interact with it easily. It focuses on core programming concepts such as:

Variables and data types

Conditional statements

Loops

Functions

Dictionaries

This project serves as a foundation for more advanced systems like full-fledged library management software.

.

AIM OF THE PROJECT

The main aim of this project is to design and develop a simple and efficient library management system using Python.

Specific Aims:

- To automate the process of managing library books
- To minimize manual errors in record-keeping
- To understand the practical application of Python programming
- To develop a structured and user-friendly system
- To enhance logical thinking and coding skills

.

PROBLEM DEFINITION AND OBJECTIVES

Problem Definition:

In traditional libraries, book records are maintained manually in registers. This approach has several drawbacks:

Difficulty in searching for books

Chances of human errors

Time-consuming processes

Lack of proper tracking system

There is a need for a simple automated system that can efficiently manage library records.

Objectives:

The objectives of this project are:

Efficient Data Management:

Store and manage book details using Python data structures.

Book Availability Tracking:

Keep track of whether a book is available or issued.

User Interaction:

Provide a simple menu-based interface for easy use.

Error Reduction:

Reduce mistakes compared to manual systems.

Scalability:

Provide a base system that can be expanded with more features.

. **CODING**

Description of Code:

The program uses a dictionary to store book details, where:

Key = Book ID

Value = Book name and availability status

Functions are used to organize the code and improve readability

CODE:

```
# Library Book Tracker System


library = {}


def add_book():
    book_id = input("Enter Book ID: ")
    name = input("Enter Book Name: ")


    if book_id in library:
        print("Book ID already exists!")
    else:
        library[book_id] = {"name": name, "available": True}
        print("Book added successfully!")


def view_books():
```

```
    if not library:
        print("No books available.")
    else:
        print("\nLibrary Books:")
        for book_id, details in library.items():
            status = "Available" if details["available"] else "Issued"
            print(f"ID: {book_id}, Name: {details['name']}, Status: {status}")


def issue_book():
    book_id = input("Enter Book ID to issue: ")
    if book_id in library:
        if library[book_id]["available"]:
            library[book_id]["available"] = False
            print("Book issued successfully!")
        else:
            print("Book is already issued.")
    else:
        print("Book ID not found.")


def return_book():
    book_id = input("Enter Book ID to return: ")
    if book_id in library:
        if not library[book_id]["available"]:
```

```
            library[book_id]["available"] = True
            print("Book returned successfully!")
        else:
            print("Book was not issued.")
    else:
        print("Book ID not found.")

# Main Program
while True:
    print("\n====== Library Menu ======")
    print("1. Add Book")
    print("2. View Books")
    print("3. Issue Book")
    print("4. Return Book")
    print("5. Exit")

    choice = input("Enter your choice: ")

    if choice == '1':
        add_book()
    elif choice == '2':
        view_books()
    elif choice == '3':
```

```
    issue_book()
elif choice == '4':
    return_book()
elif choice == '5':
    print("Thank you for using the system!")
    break
else:
    print("Invalid choice! Please try again.")
```

.

OUTPUT SCREENS

- Output Example 1:

====== Library Menu ======

1. Add Book
2. View Books
3. Issue Book
4. Return Book
5. Exit

Enter your choice: 1

Enter Book ID: 201

Enter Book Name: Data Structures

Book added successfully!

- Output Example 2:

Enter your choice: 2

Library Books:

ID: 201, Name: Data Structures,

Status: Available

- Output Example 2:

Enter your choice: 3
Enter Book ID to issue: 201
Book issued successfully!

- Output Example 2:

Enter your choice: 4

Enter Book ID to return: 201

Book returned successfully!

.

FUTURE WORK

This project can be further enhanced by adding advanced features:

- File Handling: Save data permanently in files
- Graphical User Interface (GUI): Using Tkinter
- Login System: For admin and users
- Due Dates & Fine Calculation: For late returns
- Database Integration: Using MySQL or SQLite
- Web/App Version: Convert into web-based system

.

CONCLUSION

The Library Book Tracker System successfully demonstrates how Python programming can be used to develop a simple yet effective solution for managing library records. The system reduces manual effort, improves accuracy, and provides an easy way to track books.

This project not only strengthens programming fundamentals but also helps students understand how software solutions are applied in real-world scenarios. With further improvements, this system can be developed into a complete library management application.